

PAPER_1596_SOLAR_CONSTANT_1361

Daniel T. Murphy

$$\text{PAPER_1596} \text{ — Solar Constant (TSI)} = S = A_5^2 \cdot F \cdot D - N_{CH} \cdot SO_5 + SO_5 + SSQ + F \cdot D = 1440 - 90 + 10 + 0.57 + 0.4 = 1360.97 \text{ W/m}^2$$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Climate

Date: June 18, 2026

Status: CLOSED — 0.002% integer-primitive identity

Observation

PAPER_1209X_S556 (Climate Unified Proof Set) lists **Solar Constant (TSI)** with the integer-primitive expression:

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$$S = A_5^2 \cdot F \cdot D - N_{CH} \cdot SO_5 + SO_5 + SSQ + F \cdot D = 1440 - 90 + 10 + 0.57 + 0.4 = 1360.97 \text{ W/m}^2$$

``

Residual: **0.002%**.

UQFF Closed Identity

``

$$S = A_5^2 \cdot F \cdot D - N_{CH} \cdot SO_5 + SO_5 + SSQ + F \cdot D = 1440 - 90 + 10 + 0.57 + 0.4 = 1360.97 \text{ W/m}^2 \quad 0.002\%$$

``

The formula uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical climate measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209X_S556

- Related: PAPER_1494-1595 (other session-2026-06-18 closures)

- Calculator dispatch: `calculate_paradox({"paradox": "solar_constant_1361"})`

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